DBA5103: Operations Research and Analytics **Semester I, 2026/2027, NUS**

Assignment 1: Due on Sep.6, 2026

The solution is for your own reference only. **Do not CIRCULATE.**

1. (10') Consider the production problem of n products labeled by $j = 1, 2, ..., n$. The unit selling price for the j-th product is denoted by p_j. All products can be produced from m raw materials. The production of product j requires a_{ij} units of raw material i, $i = 1, ..., m$. The i-th raw material can be procured at a unit price c_i. The company has a total budget of B that can be used to procure the raw materials. Naturally, we assume that all the parameters here are nonnegative. The problem is to find the production plan that maximizes the total profit (revenue from selling the products minus the procurement cost).

   (a) (7') Let x_j be the production quantities of product j, $j = 1, ..., n$ and y_i be the procurement quantities of raw material i, $i = 1, ..., m$.

   (i) Formulate a linear programming model for the problem.

   (ii) Let

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

   denote the bill-of-material matrix and

$$\boldsymbol{p} = \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}, \ \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \ \boldsymbol{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{pmatrix}, \ \boldsymbol{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}.$$

   Express your model in matrix form.

   (b) (3') Formulate a model for the problem without introducing the procurement decisions y_i.

Solution:

(a) With the decisions variables x_j and y_i, we can write the model as

$$\begin{array}{ll} \max & \sum_{j=1}^{n} p_j x_j - \sum_{i=1}^{m} c_i y_i \\ s.t. & \sum_{j=1}^{n} a_{ij} x_j = y_i, \ i = 1, ..., m \\ & \sum_{i=1}^{m} c_i y_i \le B \\ & x_j \ge 0, j = 1, ..., n, y_i \ge 0, i = 1, ..., m. \end{array}$$

Using matrix notation, we can equivalently write the above formulation as

$$\begin{array}{ll} \max & \boldsymbol{p}'\boldsymbol{x} - \boldsymbol{c}'\boldsymbol{y} \\ s.t. & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{y} \\ & \boldsymbol{c}'\boldsymbol{y} \le B \\ & \boldsymbol{x} \ge \boldsymbol{0}, \boldsymbol{y} \ge \boldsymbol{0}. \end{array}$$

Note: the constraint $\boldsymbol{y} \geq \mathbf{0}$ can be omitted since all the parameters are nonnegative, $\boldsymbol{x} \geq \mathbf{0}$ implies that $\boldsymbol{y} \geq \mathbf{0}$ as well.

(b) Observe that the total procurement cost incurred from producing one unit of product j is simply: $\sum_{i=1}^{m} c_i a_{ij}$ and the total cost from the production is: $\sum_{j=1}^{n} \left(\sum_{i=1}^{m} c_i a_{ij}\right) x_j$. Therefore, we have

$$\begin{array}{ll} \max & \sum_{j=1}^{n}(p_j - \sum_{i=1}^{m} c_i a_{ij})x_j \\ s.t. & \sum_{j=1}^{n}\left(\sum_{i=1}^{m} c_i a_{ij}\right) x_j \leq B \\ & x_j \geq 0, j = 1, ..., n. \end{array}$$

Using matrix notation, this becomes quite clear by directly eliminating the variables $\boldsymbol{y}$ in the problem using the constraints $\boldsymbol{Ax} = \boldsymbol{y}$:

$$\begin{array}{ll} \max & (\boldsymbol{p}' - \boldsymbol{c}'\boldsymbol{A})\boldsymbol{x} \\ s.t. & \boldsymbol{c}'\boldsymbol{Ax} \leq B \\ & \boldsymbol{x} \geq \mathbf{0}. \end{array}$$

Note: we have omitted the constraint $\boldsymbol{y} \geq \mathbf{0}$ in the above formulation. Adding the constraints $\boldsymbol{Ax} \geq \mathbf{0}$ is correct as well.

2. **Matrix Completion Problem:** (10') Consider the following movie-ratings matrix collected from 3 users for 5 movies. In this matrix, 1 means the user likes the movie while 0 means the user dislikes the movie. The missing entries indicate that the user did not watch the particular movie. The problem is to fill in the missing entries so as to predict whether the user will like an unwatched movie. Clearly,

	A Beautiful Mind	**Aquaman**	**Titanic**	**Black Panther**	**Schindler's List**
User 1	1	0	1		
User 2			0		
User 3	1			0	1

without any restriction on the matrix, this is an impossible task as one can arbitrarily fill in either 0 or 1 in the missing entries. In practice, however, as in the compressed sensing example discussed in class, the matrix is known to be structured in the sense that it is *low-rank*. We will explore this concept in this exercise and see how it is helpful in matrix completion problem.

Formally, the *rank* of a matrix A is the maximal number of linearly independent columns of A, which is also identical to the maximal number of linearly independent rows of A.

(a) (2') Compute the rank of following matrix:

$$\boldsymbol{A} = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}.$$

(b) (4') Consider two nonzero vectors

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} \in \Re^m, \text{ and } \boldsymbol{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \in \Re^n.$$

What is the rank of the matrix $\boldsymbol{A} = \boldsymbol{xy}'$? Explain your reasoning. In fact, the converse of your conclusion here also holds: any matrix $\boldsymbol{A}$ that has the rank of your answer in (b) can be expressed as $\boldsymbol{A} = \boldsymbol{xy}'$ for some nonzero vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ (you can think about why; this is not required to be part your answer for part (b)).

(c) (4') Consider the matrix

$$\boldsymbol{A} = \begin{pmatrix} 1 & 0 & 1 & ? & ? \\ ? & ? & 0 & ? & ? \\ 1 & ? & ? & 0 & 1 \end{pmatrix}$$

in the moving rating example above. Suppose we know the matrix $\boldsymbol{A}$ has the rank of your answer in part (b), or equivalently $\boldsymbol{A} = \boldsymbol{x}\boldsymbol{y}'$ for some nonzero (but unknown) vectors $\boldsymbol{x}$ and $\boldsymbol{y}$. Complete the matrix $\boldsymbol{A}$ and explain your reasoning.

From the exercise in part (c), we should see that once we know the rank information of a matrix, then sometimes we can exactly recover the matrix given only a sample of its entries (in our example, we only observe less than 50% of the entries). The structural knowledge that $\boldsymbol{A} = \boldsymbol{x}\boldsymbol{y}'$ for some $\boldsymbol{x}$ and $\boldsymbol{y}$ also has practical interpretations. For example, we can interpret $y_j, j = 1, ..., n$ as some hidden attribute of the movie j (say, whether movie j has won some award or not), and interpret $x_i, i = 1, ..., m$ as customer i's taste for the hidden attribute. Such structure then presumes that customer i's rating for movie j follows the formula $x_i y_j$. Our example here is only a very simplified version of the matrix completion problem. Interested students are referred to Candès and Recht (2009) for more general discussion of the problem.

Solution:

(a) There are only two linearly independent rows: the first row and the second row are linearly independent with the third row being a linear combination of the first two rows. Hence, the rank of $\boldsymbol{A}$ is two. One can also check that there are only two linearly independent columns among the four columns.

(b) Using matrix multiplication, we have

$$\boldsymbol{A} = \begin{pmatrix} x_1y_1 & x_1y_2 & \dots & x_1y_n \\ x_2y_1 & x_2y_2 & \dots & x_2y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_my_1 & x_my_2 & \dots & x_my_n \end{pmatrix}$$

Observe that every column vector is of the form

$$x_i \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \ i = 1, ..., m.$$

Hence, they are scalar multiplicative of each other, and note that they cannot be all zero since $\boldsymbol{x}$ and $\boldsymbol{y}$ are assumed to be nonzero. As a result, the rank of $\boldsymbol{A}$ must be one.

Conversely, if a matrix $\boldsymbol{A}$ has rank one, then all the rows must be linearly dependent and can be expressed as a scalar multiplied by some vector. That is, there exists x_i and $\boldsymbol{y} = (y_1, ..., y_n)'$ such that row i is $x_i(y_1, ..., y_n)$, and we have

$$\boldsymbol{A} = \begin{pmatrix} x_1y_1 & x_1y_2 & \dots & x_1y_n \\ x_2y_1 & x_2y_2 & \dots & x_2y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_my_1 & x_my_2 & \dots & x_my_n \end{pmatrix} = \boldsymbol{x}\boldsymbol{y}'.$$

(c) Using part (b), we can express the matrix $\boldsymbol{A}$ as

$$\boldsymbol{A} = \begin{pmatrix} x_1y_1 & x_1y_2 & x_1y_3 & x_1y_4 & x_1y_5 \\ x_2y_1 & x_2y_2 & x_2y_3 & x_2y_4 & x_2y_5 \\ x_3y_1 & x_3y_2 & x_3y_3 & x_3y_4 & x_3y_5 \end{pmatrix}.$$

Note first from the third column that we know $x_1y_3 = 1$ and $x_2y_3 = 0$. This implies $y_3 \neq 0$ and $x_2 = 0$. Note further from the fist column that we know $x_1y_1 = 1$ and $x_3y_1 = 1$. This implies $x_1 = x_3 \neq 0$. Consequently, the first row and the third row of $\boldsymbol{A}$ must have the same value while the second row must be all zero, and it follows that

$$\boldsymbol{A} = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix}.$$

3. (5') Recall the definition that a *polyhedron* is a set that can be described by a finite number of linear equalities and inequalities, i.e., it is the intersection of hyperplanes and halfspaces. Consider each of the following feasible sets and answer the corresponding questions.

(a) (2') Let a be a nonnegative parameter. The set of all $(x, y) \in \Re^2$ satisfying the constraints:

$$\begin{aligned} a^2x + \sqrt{a}y &\leq 1, \\ x &\geq 0, \\ y &\geq 0. \end{aligned}$$

Is it a polyhedron? Draw a sketch of the feasible set for $a = 1$.

(b) (1') The set of all $(x, y) \in \Re^2$ satisfying the constraints:

$$\begin{aligned} x^2 + y^2 &\leq 1, \\ x &\geq 0, \\ y &\geq 0. \end{aligned}$$

Draw a sketch of the feasible set. Is it a polyhedron?

(c) (1') $\{x \in \Re | x^2 - 8x + 15 \leq 0\}$. Is it a polyhedron?

(d) (1') How about $\{x \in \Re | x^2 - 8x + 15 \geq 0\}$?

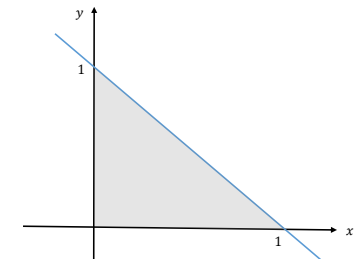
Solution:

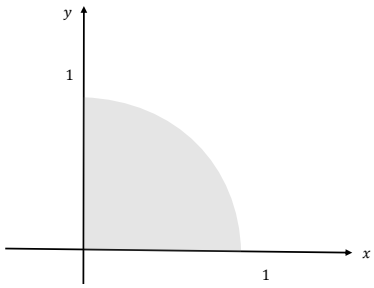
(a) Note that a is a parameter and the nonlinear terms a^2, $\sqrt{a}$ do not affect the fact that the constraints are linear in (x, y). Hence it is a polyhedron. For example, when $a = 1$, it is shown in the figure below.

(b) The constraint $x^2 + y^2 \leq 1$ cannot be expressed as linear constraints. This is not a polyhedron. Its sketch is shown in the figure.

(c) The constraint can be equivalently expressed as $x \geq 3$ and $x \leq 5$. Hence, it is a polyhedron.

(d) The constraint can be equivalently expressed as $x \geq 3$ or $x \leq 5$. It is not an intersection of halfspaces and hence not a polyhedron.





4. **Manpower Scheduling:** (10') Ricbert is the chief bank teller of the Gringotts Wizarding Bank. He is responsible to schedule the staffing of the clerks for the bank. The bank opens from 8 A.M. until midnight. Ricbert has monitored the customers visiting the bank at various times of the day, and determined that the following number of clerks are required:

   Two types of clerks can be hired: full-time and part-time. The full-time clerks work for 8 consecutive hours in any of the following shifts: morning (8 A.M.-4 P.M.), afternoon (noon-8 P.M.), and evening (4 P.M.-midnight). Full-time clerks are paid $40 per hour.

   Part-time clerks can be hired to work any of the four shifts listed in the above table (i.e., 8 A.M.-noon, noon-4 P.M., 4 P.M.-8 P.M. and 8 P.M.-midnight). Part-time clerks are paid $30 per hour.

   An additional requirement is that during every time period, the number of full-time clerks on duty cannot be strictly less than the number of part-time clerks on duty.

   Ricbert would like to determine how many full-time and how many part-time clerks should work each shift to meet all the above requirements at the minimum possible cost.

   Formulate the problem as a linear programming problem (ignoring the constraints that the number of clerks are required to be integers).

   Solution:

   Let x_1, x_2, x_3 be the number of full-time clerks scheduled in the morning, afternoon and evening shifts respectively and y_1, y_2, y_3, y_4 be the number of part-time clerks scheduled in the four time periods. We

Time of Day	**Minimum Number of Clerks Required to Be on Duty**
8 A.M.-noon	4
Noon-4 P.M.	8
4 P.M.-8 P.M.	10
8 P.M.-midnight	6

Table 1.1: Minimum Number of Clerks Required for Each Time Period

can formulate the problem as follows:

$$\begin{aligned} \min \quad & 320(x_1+x_2+x_3)+120(y_1+y_2+y_3+y_4) \\ \text{s.t} \quad & x_1+y_1 \geq 4 \\ & x_1+x_2+y_2 \geq 8 \\ & x_2+x_3+y_3 \geq 10 \\ & x_3+y_4 \geq 6 \\ & x_1 \geq y_1 \\ & x_1+x_2 \geq y_2 \\ & x_2+x_3 \geq y_3 \\ & x_3 \geq y_4 \\ & x_1,x_2,x_3,y_1,y_2,y_3,y_4 \geq 0 \end{aligned}$$

5. (5') Consider the problem

$$\begin{aligned} \min \quad & 2x_1+3|x_2-10| \\ s.t. \quad & |x_1+2|+|x_2| \leq 5, \end{aligned}$$

and reformulate it as a linear programming problem.

Solution: Note that $|x_1+2|+|x_2| = \max\{x_1+2+x_2, x_1+2-x_2, -x_1-2+x_2, -x_1-2-x_2\}$, and we have

$$\begin{aligned} \min \quad & 2x_1+3x_3 \\ s.t. \quad & x_1+2+x_2 \leq 5, \\ & x_1+2-x_2 \leq 5, \\ & -x_1-2+x_2 \leq 5, \\ & -x_1-2-x_2 \leq 5, \\ & x_2-10 \leq x_3, \\ & -x_2+10 \leq x_3. \end{aligned}$$

An alternative (and better) formulation is

$$\begin{aligned} \min \quad & 2x_1+3x_3 \\ s.t. \quad & y_1+y_2 \leq 5, \\ & x_1+2 \leq y_1, \\ & -x_1-2 \leq y_1, \\ & x_2 \leq y_2, \\ & -x_2 \leq y_2, \\ & x_2-10 \leq x_3, \\ & -x_2+10 \leq x_3. \end{aligned}$$

6. **Investment under Taxation:** (10') An investor has a portfolio of n different stocks. He has bought s_i shares of stock i at price $p_i, i = 1, ..., n$. The current price of one share of stock i is q_i. The investor expects that the price of one share of stock i in one year will be r_i. If he sells shares, the investor pays transaction costs at the rate of 1% of the amount transacted. In addition, for each stock, the investor pays taxes at the rate of 30% on capital gains (there is no tax if there is no capital gain). For example, suppose that the investor sells $1,000$ shares of a stock at \$50 per share. He has bought these shares at \$30 per share. Upon selling, he receives $1,000 \times 50 = \$50,000$. However, he owes $0.30 \times (50,000 - 30,000) = \$6,000$ on capital gain taxes and $0.01 \times 50,000 = \$500$ on transaction costs. So, by selling $1,000$ shares of this stock he nets $50,000 - 6,000 - 500 = \$43,500$. On the other hand, if the current price of the stock is \$20 per share instead of \$50. Then, there is no capital gain on this stock and hence no tax to be paid. In this case, by selling 1,000 shares, he nets $20,000 - 0.01 \times 20,000 = \$19,800$.

   (a) (6') Formulate the problem of selecting how many shares the investor needs to sell in order to raise an amount of money at least K, net of capital gain taxes and transaction costs, while maximizing the expected value of his (remaining) portfolio next year.

   (b) (4') Using the data for the portfolio in investment.csv, solve the problem for $K = \$9,000$ and attach your code.

   Solution: (a) Let x_i be the number of shares of stock i to be sold. The amount of money raised from selling x_i shares of stock i is then $q_i x_i - 0.3 \times \max\{q_i - p_i, 0\}x_i - 0.01 q_i x_i)$. The LP formulation of the problem is then

$$\begin{array}{ll} \max & \sum_{i=1}^{n} r_i(s_i - x_i) \\ s.t. & \sum_{i=1}^{n}(q_i x_i - 0.3\max\{q_i - p_i, 0\}x_i - 0.01 x_i q_i) \geq K \\ & 0 \leq x_i \leq s_i, \forall i = 1, ..., n \end{array}$$

   (b) Please refer to Assignment_1_Code.ipynb for the code. The maximum expected value of the portfolio in the next year is \$15356.8 and the optimal selling strategy is provided in the table.

	S1	S2	S3	S4	S5
Number of shares	1,000	1,000	1,000	118.762	0

References

Candès, E. J. and B. Recht (2009). Exact matrix completion via convex optimization. *Foundations of Computational mathematics 9*(6), 717–772.